👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.NOS.8

# Compare and Order Integers and Rational Numbers

Lesson 7-4 · Teacher Copy

**Name:** _____ **Date:** _____ **Class:** _____

## My Notes

**Level 3 Enrichment**

🎯 **TODAY'S OBJECTIVES**

**Content Objective: I can compare and order integers using a number line.**
**Language Objective: I can explain my reasoning using the words compare, order, greater than, and less than.**

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

**Number line** · **Compare** · **Order** · **Greater than** · **Less than** · **Negative integer** · **Integer** · **Inequality**

*Level 3: try the blanks from memory first, then check the bank.*

1. To plot integers in order, place them on a _____.

2. To decide which number is larger or smaller is to _____ them.

3. To arrange numbers from least to greatest or greatest to least is to _____ them.

4. On a number line, a number farther to the right is _____ another number.

5. On a number line, a number farther to the left is _____ another number.

6. An integer less than zero, like –6, is a _____.

7. A whole number or its opposite is an _____.

8. A statement that compares numbers with < or > is an _____.

## Write About the Math

**How do you use integer comparison to find the coldest and warmest days?**
*¿Cómo usas la comparación de enteros para hallar el día más frío y el más caluroso?*

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Number line** — *Recta numérica*
- ☐ **Compare** — *Comparar*
- ☐ **Order** — *Ordenar*
- ☐ **Greater than** — *Mayor que*
- ☐ **Less than** — *Menor que*

**Model:** *-3 is less than 2 because on the number line ___.* — Student explains that -3 is farther left on the number line than 2, and numbers decrease moving left.

## Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **The coldest day was ___ at ___°F because ___. The warmest day was ___ at ___°F. I ordered them from coldest to warmest by ___.**
- **My answer is ___ because ___.**
  *Mi respuesta es ___ porque ___.*

## Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is ___.**
  *Mi afirmación es ___.*
- **My evidence is ___, so ___.**
  *Mi evidencia es ___, entonces ___.*

**Explain**

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

## Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

## Answer Key & Teacher Guide

1. **Notes 1:** Number line
2. **Notes 2:** Compare
3. **Notes 3:** Order
4. **Notes 4:** Greater than
5. **Notes 5:** Less than
6. **Notes 6:** Negative integer
7. **Notes 7:** Integer

8. **Notes 8:** Inequality
9. **Watch for this mistake:** *A common mistake in Compare and Order Integers is thinking that a negative number with a bigger digit is the bigger value — for example, believing -15 is greater than -8 because 15 is greater than 8. It's the opposite for negatives: the number farther to the left on the number line (farther from zero) is always the LESSER value, so -15 < -8. Before you submit, check your comparison against the number line, not just the digits.*
10. **Try It 1:** D. -4 m — *Closer to zero means greater for negatives. -4 is only 4 below the surface, so -4 is the greatest of -9, -4, and -15.*
11. **Try It 2:** D. < — *-8 is farther left (deeper) than -3, so -8 is less than -3. The symbol is <.*